Q1 Real-time feature extraction pipeline for pedestrian & vehicle detection in autonomous driving (night-time & rainy conditions)

1. PROBLEM & TECHNIQUE SELECTION

The application demands **robustness to illumination degradation** (rain, night), **real-time speed** (≥25 fps), and **repeatability across scale** for object matching. A single detector fails; a *combined cascade pipeline* is chosen: **Scharr edges → FAST interest points → ORB descriptors**, enriched with a **multi-scale Gaussian pyramid**.

2. PIPELINE

Input Frame → Gaussian Pyramid (5 levels) → Scharr Edge Detection → FAST Keypoint Detection → ORB Descriptor Extraction → Hamming-dist Matching → Object Hypothesis

3. STAGE EXPLANATIONS

S1 Gaussian Pyramid (5 levels)

Each level is formed by convolving with a Gaussian kernel G(x,y) and sub-sampling by 2, giving images at scales σ, 2σ, 4σ, … Pedestrians at 5 m and 40 m appear at different pyramid levels, enabling **scale-invariant** detection. Also smooths rain-streak noise before gradient computation.

S2 Scharr Edge Detection

Computes gradient magnitudes and orientations on each pyramid level. Sobel uses a 3×3 kernel with weights [1,2,1] / [-1,0,1], which has poor rotational isotropy. Scharr uses optimised weights that minimise rotational error, critical for slanted edges of vehicles in perspective. Canny adds hysteresis but is slower (2× more computation).

S3 FAST Interest-Point Detection

Tests 16 pixels on a Bresenham circle of radius 3. A point p is a corner if ≥9 contiguous pixels are all brighter or all darker than p by threshold t. FAST achieves ~100 fps vs. Harris at ~20 fps, critical for real-time. Non-maximum suppression removes duplicates using Harris score as tie-breaker. Shi-Tomasi improves Harris corner quality but FAST+NMS is the speed winner here.

S4 ORB Descriptor Extraction

ORB = Oriented FAST + Rotated BRIEF. It uses the intensity centroid to assign a canonical orientation θ to each keypoint, then applies steered BRIEF: 256 binary comparisons of smoothed pixel pairs in a rotated patch. The descriptor is a 256-bit string, enabling **Hamming-distance matching in O(n/64)** with bitwise XOR. Unlike SIFT (128-D float), ORB is 64× more memory efficient and patent-free.

S5 Hamming-Distance Matching + RANSAC

Match descriptors from current and reference frames; prune with RANSAC to find a consistent homography, confirming vehicle/pedestrian candidates.

4. MATHEMATICAL PRINCIPLES

SCHARR GRADIENT MAGNITUDE & ORIENTATION

$$G_x = \begin{bmatrix} -3 & 0 & +3 \\ -10 & 0 & +10 \\ -3 & 0 & +3 \end{bmatrix} * I$$

$$G_y = \begin{bmatrix} -3 & -10 & -3 \\ 0 & 0 & 0 \\ +3 & +10 & +3 \end{bmatrix} * I$$

$$M(x,y) = \sqrt{G_x^2 + G_y^2} \quad \theta(x,y) = \operatorname{atan2}(G_y, G_x)$$

ORB ORIENTATION VIA INTENSITY CENTROID

$$m_{00} = \sum I(x,y), \quad m_{10} = \sum x \cdot I(x,y), \quad m_{01} = \sum y \cdot I(x,y)$$

$$C = \left(\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}}\right), \quad \theta = \operatorname{atan2}(m_{01}, m_{10})$$

HARRIS CORNER SCORE (FOR NMS PRIORITISATION)

$$R = \det(M) - k \cdot (\operatorname{trace}(M))^2, \quad k \approx 0.04$$

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} \text{ (summed over patch)}$$

GAUSSIAN PYRAMID LEVEL L

$$G_l(x,y) = \sum_{u,v} w(u,v) \cdot G_{l-1}(2x+u, 2y+v)$$

$$w = \frac{1}{256} \cdot [1,4,6,4,1]^T [1,4,6,4,1] \text{ (binomial approximation to Gaussian)}$$

BRIEF BINARY TEST

$$\tau(p; x, y) = \begin{cases} 1 & \text{if } I(p+x) < I(p+y) \\ 0 & \text{else} \end{cases}$$

$$f_n(p) = \sum_{1 \le i \le n} 2^{(i-1)} \cdot \tau(p; x_i, y_i), \quad n = 256$$

5. JUSTIFICATION OF TECHNIQUE SELECTION

DETECTOR/ DESCRIPTOR	WHY CHOSEN	COMPETITOR REJECTED
Scharr	Better rotational isotropy than Sobel; same speed; crucial for angled vehicle edges	Canny: slower, hysteresis tuning difficult in dynamic night lighting
FAST	~100 fps on CPU; repeatable under illumination shifts	Harris: ~20 fps; Shi-Tomasi: good quality but lacks speed
ORB	Binary, rotation-invariant, scale-invariant (via pyramid), free	SIFT: too slow (patent expired but heavy); BRIEF: not rotation-invariant
Pyramid	Multi-scale without re-running full detector; O(4/3 n) cost	DOG: more accurate but costlier

6. ADVANTAGES & LIMITATIONS

ADVANTAGES

- Real-time capable (>30 fps on embedded GPU)
- Rotation & scale invariant
- Robust to mild illumination change (Scharr isotropy)
- Memory-efficient (256-bit ORB vs 512-byte SIFT)
- Patent-free (ORB)

LIMITATIONS

- FAST not inherently scale-invariant (mitigated by pyramid)
- Binary descriptors degrade with heavy blur (rain drops)
- Rain streaks at night can create false edges
- RANSAC non-deterministic — may fail under occlusion

Q2 Interactive segmentation pipeline for 3-D MRI tumour extraction with intensity inhomogeneity and weak boundaries

1. PROBLEM & TECHNIQUE SELECTION

MRI volumes exhibit *bias-field inhomogeneity* (slow intensity drift), *weak boundaries*, and *topology sensitivity*. A single method fails: Watershed alone over-segments; Snakes leak through weak edges; Graph-Cut without initialisation is underconstrained. The chosen pipeline: **Active Contours (Level-Set) → Watershed (marker-controlled) → Graph-Cut (with user seeds)**.

2. PIPELINE

Noisy 3-D MRI Volume → Bias-Field Correction (N4) → Level-Set Active Contour → Marker-Ctrl Watershed → Graph-Cut Refinement → Volume Measurement

3. STAGE EXPLANATIONS

S1 Bias-Field Correction (N4ITK)

Estimates and removes the slow-varying multiplicative bias field B(x) where observed intensity I(x) = B(x)·T(x) + n(x), T(x) = true tissue intensity. Ensures Watershed thresholds are meaningful.

S2 Level-Set Active Contour (Chan-Vese)

Represents the evolving boundary as the zero-level of a signed distance function φ(x,t). Energy functional $E = \lambda_1 \int (I - c_1)^2 d\Omega_1 + \lambda_2 \int (I - c_2)^2 d\Omega_2 + \mu \cdot |\partial\Omega|$. User places an initial ellipsoid seed around the tumour; the contour evolves via gradient descent on E, stopping at the weak boundary where intensity gradient is small. Handles topology changes (tumour splitting) unlike parametric Snakes.

S3 Marker-Controlled Watershed

The level-set result provides foreground markers; user scribbles on background provide background markers. The gradient magnitude image is treated as a topographic map; flooding from markers prevents over-segmentation. Each catchment basin corresponds to a region. Stops leakage across weak boundaries by confining flooding to marker-controlled basins.

S4 Graph-Cut Refinement

Construct graph G=(V,E): each voxel is a node. Terminal edges (t-links) encode P(voxel ∈ tumour) from appearance model; neighbourhood edges (n-links) encode boundary penalty. Min-cut = max-flow (Ford-Fulkerson) finds globally optimal binary labelling, refining the boundary at pixel level. User scribbles add hard constraints (seeds) on t-link weights.

4. MATHEMATICAL PRINCIPLES

CHAN-VESE LEVEL-SET ENERGY

$$E(\varphi, c_1, c_2) = \mu \int |\nabla H(\varphi)| dx + \lambda_1 \int (I - c_1)^2 H(\varphi) dx + \lambda_2 \int (I - c_2)^2 (1 - H(\varphi)) dx$$

$$\partial\varphi/\partial t = \delta(\varphi)[\mu \cdot \operatorname{div}(\nabla\varphi/|\nabla\varphi|) - \lambda_1 (I - c_1)^2 + \lambda_2 (I - c_2)^2]$$

GRAPH-CUT MIN-CUT / MAX-FLOW

$$E(L) = \sum_p \theta_p(L_p) + \sum_{\{p,q\} \in N} \theta_{pq}(L_p, L_q)$$

$$\theta_p(0) = -\log P(I_p \mid \text{background}), \quad \theta_p(1) = -\log P(I_p \mid \text{tumour})$$

$$\theta_{pq}(L_p \neq L_q) = \exp(-(I_p - I_q)^2 / 2\sigma^2) / \operatorname{dist}(p,q)$$

5. JUSTIFICATION & ORDER

Level-Set first provides a coarse, topologically flexible initialisation that handles the tumour's irregular shape. Watershed then exploits this to avoid the classic *over-segmentation problem*. Graph-Cut last delivers **globally optimal refinement** with user-interactive constraints, ensuring precise boundary delineation for subsequent volumetry.

ADVANTAGES

- Handles topology changes (Level-Set)
- Leakage prevention (markers in Watershed)
- Global optimality (Graph-Cut)
- Interactive via seeds/scribbles
- Topologically correct masks for shape analysis

LIMITATIONS

- Level-Set re-initialisation needed to avoid numerics
- Watershed still may over-segment near noise
- Graph-Cut: memory O(V+E) — large 3-D volumes costly
- User seed quality critically affects result

Q3 Hybrid anomaly-detection scheme for high-dimensional IoT sensor streams

1. PROBLEM & TECHNIQUE SELECTION

IoT streams are high-dimensional, *largely unlabelled*, subject to *concept drift*, and contain very few confirmed anomaly labels. A two-stage hybrid is chosen: **DBSCAN (unsupervised clustering)** followed by **One-Class SVM (semi-supervised boundary)**.

2. PIPELINE

IoT Stream Window → Feature Extraction (PCA) → DBSCAN Clustering → Normal-Class Model (OC-SVM) → Anomaly Decision → Online Model Update

3. STAGE EXPLANATIONS

S1 Feature Extraction & Windowing

Sliding window of W samples; compute time-domain (mean, variance, RMS) and frequency-domain (top-k FFT coefficients) features. Apply Incremental PCA to project to d-dimensional subspace (d≪original), removing correlated sensor noise.

S2 DBSCAN Clustering

Groups unlabelled normal data without specifying k. Parameters: ε (neighbourhood radius), MinPts. Core points have ≥ MinPts in ε-ball. Points not reachable from any core point are labelled as noise — these become initial anomaly seeds. DBSCAN's density-based criterion makes it robust to outliers and arbitrary cluster shapes, unlike K-Means which forces spherical clusters.

S3 One-Class SVM

Trained on normal cluster members. Finds a hyperplane in kernel feature space that separates normal data from the origin with maximum margin ρ. Decision function: f(x) = sign(Σ α_i K(x_i,x) − ρ). A new sample with f(x)<0 is declared anomalous. RBF kernel used for non-linear boundary. ν hyperparameter upper-bounds the fraction of outliers in training set.

S4 Online Decision & Model Update

Sliding window re-runs DBSCAN every T seconds. Confirmed normal points (high confidence) are added to OC-SVM support vector set via incremental update. Concept drift is handled by exponential forgetting: old support vectors are down-weighted over time.

4. MATHEMATICAL PRINCIPLES

DBSCAN CORE POINT CRITERION

$$\text{Core}(p) \text{ iff } |\{q : \text{dist}(p,q) \le \varepsilon\}| \ge \text{MinPts}$$

Noise: points not density-reachable from any core point → anomaly candidates

ONE-CLASS SVM OPTIMISATION

$$\min_{w,\xi,\rho} \tfrac{1}{2}\|w\|^2 - \rho + \frac{1}{\nu n}\sum \xi_i$$

$$\text{s.t. } w\cdot\Phi(x_i) \ge \rho - \xi_i,\ \xi_i \ge 0$$

$$\text{Anomaly: } f(x) = \text{sgn}(w\cdot\Phi(x) - \rho) = -1$$

ADVANTAGES

- No k needed for clustering (DBSCAN)
- Works with very few labelled anomalies
- Non-parametric cluster shapes
- Kernel OC-SVM captures complex normal manifold
- Online updating for concept drift

LIMITATIONS

- DBSCAN scales as O(n log n) with k-d tree; slow for very high n
- ε and MinPts require domain knowledge
- OC-SVM kernel matrix is O(n^2) memory
- Incremental SVM update is complex

Q4 Dimensionality-reduction + classification pipeline for hyperspectral image patches with change-detection

1. PROBLEM & TECHNIQUE SELECTION

Hyperspectral images have hundreds of spectral bands (200–400). Direct classification suffers from the *Hughes phenomenon* (curse of dimensionality). Change detection requires efficient feature comparison across time. Selected: **PCA (reduction) → k-NN classifier → Background Modelling + Optical Flow (change detection)**.

2. PIPELINE

Hyperspectral Patches (D bands) → PCA (D → d) → k-NN Classifier → Label Map → BG Modelling (GMM/MOG2) → Optical Flow (Farneback) → Change Map

3. STAGE EXPLANATIONS

S1 PCA — Dimensionality Reduction

Computes covariance matrix C = (1/n)X^TX of centred spectral vectors. Eigendecomposition gives principal components; top d eigenvectors (explaining 99% variance) form projection matrix W. Projected data z = W^Tx has d≪D. This removes spectral redundancy, suppresses sensor noise, and speeds subsequent classifier training. PCA is preferred over t-SNE here because t-SNE is non-linear and non-parametric (cannot project new test patches).

S2 k-NN Classifier (Non-parametric)

In the d-dimensional PCA space, classify each patch by majority vote of its k nearest labelled neighbours using Euclidean distance. Non-parametric: no assumption on class distribution, handles arbitrary class shapes. In low-dimensional PCA space, k-NN is efficient with k-d trees (O(d log n) per query). Output is a per-pixel land-cover label map.

S3 Background Modelling (MOG2)

Models each pixel's spectral history in the PCA-reduced space as a mixture of Gaussians: B(x,t) = Σ w_k N(μ_k,Σ_k). A new frame pixel is foreground (changed) if its Mahalanobis distance to all background Gaussians exceeds a threshold. The reduced dimensionality makes Gaussian covariance estimation far more reliable (fewer parameters).

S4 Dense Optical Flow (Farneback)

Estimates the 2-D velocity field u(x,y) between consecutive PCA-compressed images using polynomial expansion of local image structure. Flow vectors on foreground pixels from the background model indicate *motion direction and speed* of changed objects, enabling discrimination of genuine change from illumination shift.

4. MATHEMATICAL PRINCIPLES

PCA — EIGENVALUE PROBLEM

$$C\cdot v_k = \lambda_k \cdot v_k,\ C = (1/n)X^T X$$

$$W = [v_1, v_2, \ldots, vd],\ z = W^T(x - \bar{\mu})$$

$$\text{Variance retained} = \Sigma_{i=1}^{d} \lambda_i \ /\ \Sigma_{i=1}^{D} \lambda_i \ge 0.99$$

K-NN DECISION RULE

$$\hat{y}(x) = \text{argmax_c } |\{x_n : \text{rank}(x,x_n) \le k \text{ and } y_n = c\}|$$

$$\text{Bayes error bound: } \varepsilon^*_{kNN} \le 2\cdot\varepsilon^*(1 - \varepsilon^*) \text{ as } k\to\infty$$

OPTICAL FLOW — FARNEBACK BRIGHTNESS CONSTANCY

$$I(x,y,t) = I(x+u, y+v, t+1)$$

$$I_x u + I_y v + I_t = 0 \text{ (linearised via Taylor expansion)}$$

$$\text{Farneback: local polynomial } f(x) \approx x^T A_x\, x + b_x^T x + c_x$$

ADVANTAGES

- Solves Hughes phenomenon via PCA
- k-NN is flexible, no distributional assumptions
- Dense optical flow captures motion field
- Near-real-time (PCA projection is fast)
- PCA noise filtering improves MOG2 reliability

LIMITATIONS

- PCA discards discriminative non-linear structure (LDA better for class separation)
- k-NN slow at test time for large databases
- Optical flow assumption (brightness constancy) violated under illumination change
- t-SNE cannot map new samples (cannot use for test)

Q5 Hybrid feature extraction for linear & textured structures in multi-resolution satellite images

⚙ 1. PROBLEM & TECHNIQUE SELECTION

Roads are thin, elongated, multi-scale; rivers have curved, blurred banks; crop patterns are periodic textures. Atmospheric noise degrades gradients. A **hybrid strategy** is required: **Image Pyramids + LoG (scale-space)**, **Gabor filters (orientation)**, **DWT (frequency decomposition)**, and **SIFT (invariant feature points)**.

→ 2. PIPELINE

Satellite Multi-band Image → Gaussian Pyramid + LoG Scale-Space → Gabor Filter Bank (8 ori × 5 scales) → DWT (3-level) → SIFT Keypoints → Feature Fusion & SVM

📋 3. STAGE EXPLANATIONS

S1 Image Pyramids + Laplacian of Gaussian (LoG)

Gaussian pyramid provides images at scales σ·2^l. LoG = ∇²G(x,y;σ) detects blobs (road intersections, crop circles) as zero-crossings. Scale-normalised LoG: σ²·∇²G. The DoG (Difference of Gaussians) approximates LoG cheaply. Responses at multiple pyramid levels locate structures at their *characteristic scale*, immune to atmospheric zoom-dependent blur.

S2 Gabor Filter Bank

Gabor kernel: g(x,y;λ,θ,σ,γ) = exp(−(x'²+γ²y'²)/2σ²)·cos(2πx'/λ). A bank with 8 orientations (0°, 22.5°, …, 157.5°) and 5 scales captures oriented structures (road edges, river banks) at each direction. Gabor is optimally localised in both space and spatial frequency (uncertainty principle). Crop rows with periodic spacing have peak Gabor response at spacing=λ.

S3 Discrete Wavelet Transform (3-level)

Decomposes the image into subbands: LL (approximation), LH (horizontal edges), HL (vertical edges), HH (diagonal details) at each level. Atmospheric noise (mostly high-frequency) is captured in HH and can be thresholded (denoising). Road textures occupy specific frequency bands, making DWT coefficients strong texture discriminators. Multi-level decomposition naturally provides multi-resolution analysis.

S4 SIFT Keypoints + Descriptors

Detects extrema in DoG scale-space; assigns orientation from gradient histogram; computes 128-D descriptor from 4×4 histograms. Scale and rotation invariant. SIFT localises road intersections, bridge endpoints, and landmark crop features that recur across different satellite passes, enabling **temporal change detection**.

∑ 4. MATHEMATICAL PRINCIPLES

SCALE-NORMALISED LOG

$$\text{LoG}_\sigma(x,y) = \sigma^2 \cdot (\partial^2 G/\partial x^2 + \partial^2 G/\partial y^2) * I(x,y)$$
$$\text{DoG}(x,y,\sigma) \approx \text{LoG}_\sigma(x,y) \text{ [Lindeberg 1994]}$$

GABOR 2-D FILTER

$$g(x,y;\theta,\lambda,\sigma) = \exp(-(x'^2+y^2y'^2)/2\sigma^2) \cdot \cos(2\pi x'/\lambda)$$
$$x' = x\cdot\cos(\theta)+y\cdot\sin(\theta),\ y' = -x\cdot\sin(\theta)+y\cdot\cos(\theta)$$

DWT 2-D SUBBAND DECOMPOSITION

$$LL = (H_{rows} \otimes H_{cols}) * I$$
$$LH = (H_{rows} \otimes G_{cols}) * I,\ HL = (G_{rows} \otimes H_{cols}) * I$$
$$HH = (G_{rows} \otimes G_{cols}) * I$$
$$H = \text{low-pass wavelet filter},\ G = \text{high-pass (detail) filter}$$

✓ 5. COMPARISON VS. CANNY+HOUGH / HARRIS

PROPERTY	HYBRID (PROPOSED)	CANNY + HOUGH	HARRIS CORNERS
Scale Invariance	**Yes** (pyramid + SIFT)	No — must re-run at each scale	Not inherently
Orientation Sensitivity	**Yes** (Gabor bank)	Detects lines but no texture	Only corners, not orientation
Texture Discrimination	**Yes** (Gabor + DWT)	No	No
Noise Resilience	**High** (DWT denoising)	Moderate (σ tuning)	Moderate
Computational Cost	Higher	Low	Low

ADVANTAGES

- Multi-scale, multi-orientation coverage
- DWT handles atmospheric noise
- SIFT enables cross-image matching
- Gabor optimal for periodic textures

LIMITATIONS

- High computational cost (Gabor bank + DWT + SIFT)
- Feature vector dimensionality can be very large (requires PCA pre-reduction)
- Gabor sensitive to frequency band selection
- SIFT not real-time without GPU

Q6 Detecting circular & elliptical objects (traffic signs, manhole covers) in real-time traffic video

⚙ 1. PROBLEM & TECHNIQUE SELECTION

Traffic signs are partially occluded, appear at varying distances, and must be detected under illumination variation. The chosen pipeline: **Contour Extraction (Suzuki-Abe) → Ellipse Fitting (RANSAC) → Randomised Hough Transform**, with multi-resolution preprocessing to handle scale variation.

→ 2. PIPELINE

Video Frame (Gaussian Pyramid) → Canny Edge Map → Suzuki-Abe Contour Ext. → Douglas-Peucker Simplification → RANSAC Ellipse Fit → RHT Circle Verification → Detected Objects

📋 3. STAGE EXPLANATIONS

S1 Multi-resolution Preprocessing

Apply Gaussian pyramid (3 levels) to detect signs at multiple distances. Canny edge detection with adaptive thresholds (Otsu-based) handles varying illumination.

S2 Suzuki-Abe Contour Extraction

Traverses the binary edge image using a border-following algorithm to output a hierarchical contour tree. Provides structured boundary representation (including parent-child nesting: outer sign boundary → inner symbol), enabling shape-based filtering. More informative than simple connected-component labelling.

S3 Douglas-Peucker Contour Simplification

Iteratively removes contour points that are within ε pixels of the line between their neighbours. Reduces contour point count by 60–80%, reducing RANSAC fitting cost. Preserves curvature maxima (corners of ellipse) which are key for fitting.

S4 RANSAC Ellipse Fitting

Repeatedly samples 5 points (minimum to fit a conic), fits an ellipse via Direct Least Squares, counts inliers within distance ε of the fitted ellipse. Robust to up to 50% outliers (occluded portions of sign). More robust than Direct Least Squares alone (which is sensitive to outliers). Returns the ellipse with maximum inlier count after N iterations.

S5 Randomised Hough Transform (RHT) Verification

Randomly selects 3 edge points, computes the unique circle through them (via circumcircle formula), accumulates (cx, cy, r) in parameter space. RHT needs only O(1) memory accumulator update vs. CHT's O(r) per edge point. Verifies whether the RANSAC ellipse is actually circular (for circular signs). Discards ellipses with too high eccentricity.

∑ 4. MATHEMATICAL PRINCIPLES

GENERAL CONIC (ELLIPSE) EQUATION — DLS FITTING

$$ax^2 + bxy + cy^2 + dx + ey + f = 0,\ \text{constraint: } 4ac - b^2 = 1 \text{ (ellipse)}$$
$$\text{DLS: minimise } ||Da||^2 \text{ s.t. } a^T Ca=1,\ D = \text{design matrix of contour points}$$
$$\rightarrow \text{generalised eigenvector problem: } D^T D\cdot a = \lambda\cdot C\cdot a$$

RANSAC ELLIPSE (MINIMUM SAMPLE = 5 POINTS)

$$N = \log(1-p) / \log(1-(1-e)^5)$$
$$p = 0.99 \text{ confidence},\ e = \text{outlier fraction}$$
$$\text{e.g. } e=0.5 \rightarrow N = 145 \text{ iterations}$$

CIRCUMCIRCLE OF 3 POINTS (RHT CIRCLE)

$$\text{Given } (x_1,y_1),(x_2,y_2),(x_3,y_3):$$
$$cx = (|P_1|^2(y_2-y_3) + |P_2|^2(y_3-y_1) + |P_3|^2(y_1-y_2)) / D$$
$$cy = (|P_1|^2(x_3-x_2) + |P_2|^2(x_1-x_3) + |P_3|^2(x_2-x_1)) / D$$
$$r = \text{distance}(C, P_1),\ D = 2|\det([P_2-P_1,\ P_3-P_1])|$$

✓ 5. TRADE-OFF ANALYSIS

METHOD	SPEED	OCCLUSION ROBUSTNESS	LOCALISATION ACCURACY
CHT (Standard)	Slow O(r·E)	Good	Moderate
RHT	Fast O(E)	Good	Good (probabilistic)
DLS Ellipse	Fast O(n)	Poor (outlier-sensitive)	High (no outliers)
RANSAC Ellipse	Moderate	Excellent	High
Suzuki Contour	Fast	Moderate	High

ADVANTAGES

- RANSAC handles 50%+ occlusion
- RHT memory-efficient vs CHT
- Contour hierarchy helps nested sign structure
- Multi-resolution handles distance variation

LIMITATIONS

- RANSAC iterations can be slow for many candidates
- DLS numerically unstable near degenerate conics
- RHT misses circles when edge density is low
- Douglas-Peucker ε tuning required

Q7 Interactive 2-D/3-D blood vessel segmentation from angiographic CT volumes

⚙ 1. PROBLEM & TECHNIQUE SELECTION

Blood vessels are thin (1–3 voxels), tubular, with intensity variation along their length, and must be separated from surrounding tissue. Pipeline: **Hessian-based Vesselness Filtering → Region Growing (seed-based) → Graph-Cut refinement**.

→ 2. PIPELINE

Angio CT Volume → Hessian Vesselness (Frangi) → Region Growing (User Seeds) → Graph-Cut Refinement → Centreline Extraction

📋 3. STAGE EXPLANATIONS

S1 — Hessian-based Vesselness Filtering (Frangi 1998)

At scale σ, compute the Hessian matrix $H(x;\sigma) = \sigma^2 \cdot [\partial^2 G/\partial x_i x_j] * I(x)$. Eigenvalues $\lambda_1 \le \lambda_2 \le \lambda_3$ of H characterise local geometry: for a bright tubular vessel, $\lambda_1 \approx 0$, $\lambda_2 < 0$, $\lambda_3 << 0$. Frangi vesselness measure: $V(x;\sigma) = \exp(-R_A^2/2\alpha^2)\cdot(1-\exp(-R_B^2/2\beta^2))\cdot(1-\exp(-S^2/2c^2))$. Applying at multiple σ values and taking max selects vessels at their characteristic scale. This *preserves thin structures* that intensity thresholding would miss.

S2 — Seeded Region Growing

User places seeds inside a vessel segment. Algorithm adds neighbouring voxels if | I(neighbour) − mean(region)| < threshold T. Grows along the vesselness-enhanced image rather than the raw intensity, preventing leakage into surrounding tissue (different Hessian signature). Threshold T adapts locally: T = k·σ_region, preventing leakage as the region grows.

S3 — Graph-Cut Refinement

Constructs a graph where t-links encode P(voxel = vessel) from the vesselness map; n-links encode intensity similarity. Min-cut separates vessel from background globally, correcting leaks and gaps from region growing. User background scribbles add hard background constraints.

S4 — Centreline Extraction

Topological thinning (skeletonisation) of the segmented binary volume extracts the vessel centreline — a topologically correct tree used for stenosis measurement and surgical planning.

∑ 4. MATHEMATICAL PRINCIPLES

FRANGI VESSELNESS MEASURE

$$R_A = |\lambda_2|/|\lambda_3| \text{ (plate vs. tube)}, \quad R_B = |\lambda_1|/\sqrt{|\lambda_2\lambda_3|} \text{ (blob)}$$
$$S = ||H||_F = \sqrt{\lambda_1^2+\lambda_2^2+\lambda_3^2} \text{ (structureness)}$$
$$V(s) = (1-\exp(-R_B^2/2\beta^2)) \cdot \exp(-R_A^2/2\alpha^2) \cdot (1-\exp(-S^2/2c^2))$$

ADAPTIVE REGION GROWING CRITERION

Add voxel q to region R iff:

$$|I(q) - \mu_R| \le k \cdot \sigma_R \text{ and } V(q) \ge \theta_{vessel}$$

μ_R = mean intensity of region, σ_R = std dev of region

ADVANTAGES

- Hessian detects thin 1–2 voxel vessels missed by thresholding
- Multi-scale vesselness handles varying vessel calibres
- Graph-Cut globally optimal — prevents topological errors
- User seeds enable interactive refinement

LIMITATIONS

- Hessian fails at vessel bifurcations (eigenvalue structure breaks down)
- Region growing can still leak at stenotic regions
- Graph-Cut memory O(V³) for 3-D — volumetric graphs are large
- Centreline extraction sensitive to holes in segmentation

Q8 Hybrid pattern-recognition scheme for fault detection in industrial vibration sensor data

⚙ 1. PROBLEM & TECHNIQUE SELECTION

Industrial vibration signals exhibit complex multi-modal distributions; fault signatures are rare and subtle. Pipeline: **K-Means++ (unsupervised grouping) → Bayesian Classifier with Mahalanobis Distance (supervised discriminant)**.

→ 2. PIPELINE

Vibration Signal Window → Feature Ext. (FFT + MFCC) → K-Means++ Clustering → Cluster Labelling (Few Labels) → Bayes + Mahal. Discriminant → Fault / Normal Decision

📋 3. STAGE EXPLANATIONS

S1 — Feature Extraction

FFT spectrum (top-k frequency amplitudes corresponding to bearing fault frequencies), RMS, kurtosis, crest factor. Kurtosis is particularly sensitive to impulsive fault signatures. These form the high-dimensional input feature vector.

S2 — K-Means++ Clustering

Initialises centroids with D^2-weighted probability (each new centroid chosen with probability proportional to squared distance from existing centroids), avoiding the worst-case clustering of K-Means random init. Minimises $\Sigma||x_i - \mu_{c(i)}||^2$. Discovers k natural operating modes (e.g. normal, worn bearing, misalignment). Better initialisation than GMM for high-dimensional vibration data.

S3 — Cluster Labelling with Scarce Labels

Expert labels a few samples from each cluster. Propagate labels to all cluster members. This is semi-supervised: we exploit the unlabelled structure found by K-Means++ to amplify scarce labels.

S4 — Bayesian Classifier + Mahalanobis Distance Discriminant

Each class k is modelled as a Gaussian $N(\mu_k, \Sigma_k)$ estimated from cluster members. The Mahalanobis distance $d_M(x,k) = \sqrt{(x-\mu_k)^T \Sigma_k^{-1} (x-\mu_k)}$ accounts for inter-feature correlations, unlike Euclidean distance. Bayesian decision: class* = argmax $P(k)\cdot N(x;\mu_k,\Sigma_k)$. A sample declared anomalous if $\min_k d_M(x,k) >$ threshold T (χ^2 distribution with d.f. = feature dimension).

∑ 4. MATHEMATICAL PRINCIPLES

K-MEANS++ INITIALISATION

$$P(x \text{ chosen as next centroid}) \propto \min_c ||x - c||^2$$
$$\text{Convergence guarantee: } E[\text{cost}] \le O(\log k) \cdot \text{optimal cost}$$

MAHALANOBIS DISTANCE & ANOMALY DECISION

$$d_M(x, \mu_k, \Sigma_k) = \sqrt{(x-\mu_k)^T \Sigma_k^{-1} (x-\mu_k)}$$
$$\text{Under normality: } d_M^2 \sim \chi^2(p),\ p = \text{feature dimension}$$
$$\text{Anomaly if: } \min_k d_M(x,k) > \chi^2_{p,\,0.99} \text{ (99th percentile)}$$

BAYESIAN DECISION RULE

$$\text{class}^* = \arg\max_k \log P(k) - \tfrac{1}{2}\log|\Sigma_k| - \tfrac{1}{2} d_M^2(x, \mu_k, \Sigma_k)$$

ADVANTAGES

- K-Means++ avoids poor local minima
- Mahalanobis handles correlated features (vibration bands correlate)
- Works with very few labelled faults
- χ^2 threshold is theoretically grounded

LIMITATIONS

- K-Means++ assumes spherical clusters (GMM better for ellipsoidal)
- Σ_k inversion ill-conditioned when features > samples
- k must be specified (requires domain knowledge of fault types)
- Sensitive to non-Gaussian distributions

Q9 Dimensionality-reduction and tracking pipeline for multi-camera surveillance video

1. PROBLEM & TECHNIQUE SELECTION

Multi-camera feature vectors are high-dimensional (appearance + spatial + temporal). Real-time tracking requires compact representations. Pipeline: **PCA (reduction) → Kernel Density Estimator (non-parametric) → MOG2 Background Subtraction + Lucas-Kanade Optical Flow**.

2. PIPELINE

Multi-cam Feature Vectors → PCA Reduction → KDE Density Est. → MOG2 BG Subtraction → Lucas-Kanade Optical Flow → Tracked Trajectories

3. STAGE EXPLANATIONS

S1 PCA Dimensionality Reduction

Projects D-dimensional appearance+spatial features to d-dimensional subspace. Preserves maximum variance, removes correlated noise from multiple cameras. Real-time: projection is a d×D matrix multiply per frame, O(dD) per vector.

S2 Kernel Density Estimator (KDE)

Non-parametric: p(x) = (1/nh^d) Σ K((x−xᵢ)/h). In the PCA space, KDE models the distribution of pedestrian appearances without assuming Gaussianity. High-density regions = common appearances (normal crowd); low-density = unusual persons. Bandwidth h chosen by Silverman's rule: h = 1.06 σ n^(-1/(d+4)). Used to flag anomalous appearances before motion analysis.

S3 MOG2 Background Subtraction

In the PCA-reduced colour/texture space, models each pixel's temporal history as a mixture of Gaussians. Foreground = pixels whose current value is improbable under background model. PCA reduction means fewer Gaussian components needed, improving speed and reducing false positives from noise.

S4 Lucas-Kanade Sparse Optical Flow

Tracks FAST keypoints in foreground regions. LK assumes brightness constancy and small motion, solving the overconstrained system ∇I·v = −I_t in a W×W window via least squares. Pyramidal implementation handles large inter-frame motion (running persons). Provides velocity vectors for each tracked foreground object, enabling trajectory construction and speed estimation.

4. MATHEMATICAL PRINCIPLES

KDE WITH GAUSSIAN KERNEL

$$\hat{p}(x) = 1/(n \cdot h^d) \cdot \Sigma_i \exp(-||x-x_i||^2/2h^2)$$

$$\text{Silverman bandwidth: } h = 1.06 \cdot \hat{\sigma} \cdot n^{-1/(d+4)}$$

LUCAS-KANADE OPTICAL FLOW (LEAST SQUARES)

$$[u, v]^T = (A^T A)^{-1} A^T b$$

$$A = [Ix_1, Iy_1; Ix_2, Iy_2; \ldots], \; b = -[It_1; It_2; \ldots]$$

$$\text{Condition: } A^T A = \Sigma[Ix^2, IxIy; IxIy, Iy^2] \text{ well-conditioned} \rightarrow \text{Harris corner}$$

ADVANTAGES

- PCA enables real-time processing
- KDE non-parametric: handles multi-modal crowds
- LK tracks individual keypoints precisely
- MOG2 adapts to slow background change (lighting)

LIMITATIONS

- PCA is linear — misses non-linear appearance manifolds (ICA or Autoencoder better)
- KDE scales as O(n) per query for n training samples
- LK assumes small motion; fails for fast-moving objects
- MOG2 slow to adapt when camera shakes

Q10 Hybrid feature extraction for dermoscopic skin lesion segmentation with specular noise

1. PROBLEM & TECHNIQUE SELECTION

Dermoscopic images have specular highlights (specular reflections from oily skin), hair artefacts, irregular lesion textures, and colour variations. A hybrid strategy: **Gaussian Scale-Space + Derivatives**, **Gabor Filters**, **DWT**, and **LBP**, followed by SVM classification.

2. PIPELINE

Dermoscopic Image → Specular Noise Removal → Gaussian Pyramid + Derivatives → Gabor Filter Bank → DWT (2-level) → LBP Texture → SVM Lesion Mask

3. STAGE EXPLANATIONS

S1 Specular Highlight Removal

Detect specular pixels (all three RGB channels saturated simultaneously, I_R>240 & I_G>240 & I_B>240). Inpaint using Telea algorithm (fast marching method). Removes the largest source of false edges.

S2 Gaussian Scale-Space + Derivatives

Multi-scale gradient magnitude, LoG (blob detector for moles/nevi), and Hessian-based ridge detector for hair artefacts. Provides *scale-invariant* structural information. Hair = elongated ridge structures → large Hessian eigenvalue ratio at fine scales.

S3 Gabor Filter Bank (6 ori × 4 scales)

Captures pigment pattern orientation in lesions. Dermoscopic lesions exhibit radial streaming (directed Gabor response) and reticular networks (lattice Gabor response). Gabor features concatenated per pixel: 6×4 = 24 values per channel.

S4 DWT (2-level Daubechies-4)

Decomposes into LH/HL/HH subbands at 2 levels. Hair and vessel structures appear in HH (diagonal detail). Lesion border irregularity is captured in LH/HL subbands. Subband energies and standard deviations form texture features. Acts as multi-resolution noise separator.

S5 LBP (Local Binary Pattern)

For each pixel, compare its intensity to P=8 circular neighbours at radius R. Encode comparisons as binary string; compute histogram over local patch. LBP is illumination-invariant (monotonic intensity transform doesn't change ordering), making it robust to the uneven dermoscope illumination. Uniform LBP (≤2 bit transitions) captures lesion microtextures.

4. MATHEMATICAL PRINCIPLES

LBP FORMULA

$$LBP_{P,R}(xc,yc) = \Sigma_{p=0}^{P-1} s(g_p - g_c) \cdot 2^p$$

$$s(x) = 1 \text{ if } x \geq 0, \text{ else } 0$$

$$\text{Uniform: at most 2 circular bit transitions} \rightarrow 58 \text{ patterns} + 1 \text{ non-uniform bin}$$

GABOR FEATURE VECTOR PER PIXEL

$$F(x,y) = [|I * g_{\theta_1,\lambda_1}|, |I * g_{\theta_1,\lambda_2}|, \ldots, |I * g_{\theta_6,\lambda_4}|]^T$$

$$\text{24-dimensional per colour channel} \rightarrow \text{72-D Gabor feature per pixel}$$

5. COMPARISON VS. SINGLE METHODS

PROPERTY	HYBRID (PROPOSED)	CANNY ONLY	HARRIS ONLY	HOUGH ONLY
Noise Resilience	**High** (DWT + specular removal)	Low	Low	Low
Texture Capture	**Yes** (LBP + Gabor)	No	No	No
Scale Invariance	**Yes** (pyramid)	No	No	Partial
Hair Artefact Handling	**Yes** (DWT HH bands)	Sees hair as edges	No	No

ADVANTAGES

- LBP illumination-invariant — robust to dermoscope lighting
- Gabor captures pigment pattern orientation
- DWT naturally denoises and multi-resolves
- Scale-space handles lesion size variation

LIMITATIONS

- High-dimensional feature vector (Gabor+DWT+LBP) → needs PCA
- Gabor parameters (λ, θ) require tuning to lesion structure
- LBP not rotation-invariant in basic form
- Specular removal inpainting can distort lesion borders

Q11 Feature pipeline for lane-marking and road-sign detection in adverse weather (fog, rain, low light)

1. PROBLEM & TECHNIQUE SELECTION

Fog degrades contrast; rain creates specular streaks; low light reduces SNR. Standard Canny edges degrade. The selected pipeline: **Phase Congruency (illumination-invariant edges) → Progressive Probabilistic Hough (line detection) → AGAST Keypoints → HOG Descriptors**.

2. PIPELINE

Frame (Adverse Weather) → Image Pyramid Dehazing → Phase Congruency → Progressive Prob. Hough Transform → AGAST Keypoints → HOG Descriptors → SVM Sign/Lane ID

3. STAGE EXPLANATIONS

S1 Image Pyramid + Dehazing

Apply dark channel prior or Retinex-based dehazing at each pyramid level to restore contrast lost in fog. Gaussian pyramid (5 levels) then provides scale-space for detecting lane markings near and far.

S2 Phase Congruency (Kovesi)

Unlike Canny which detects intensity gradient maxima, Phase Congruency detects features where Fourier components are maximally in phase: PC(x) = E(x) / (Σ_n A_n + ε). This measure is independent of absolute intensity (illumination invariant) and relative to noise level. Critically superior to Canny and LoG in fog and rain where gradients are weak but phase coherence persists at lane markings.

S3 Progressive Probabilistic Hough Transform (PPHT)

Samples a random subset of edge points; accumulates in (ρ,θ) space. When a peak exceeds threshold, traces the full line segment; removes those edge points. Terminates when remaining edge density is low. ~10× faster than standard Hough with comparable accuracy. LSD (Line Segment Detector) is gradient-based and faster still but less robust to fog-degraded gradients.

S4 AGAST (Adaptive & Generic Accelerated Segment Test)

An optimised decision tree for FAST corner detection, 2× faster than FAST. Adapts the decision tree based on observed pixel distribution, achieving ~200 fps. Critical for real-time sign detection under rain where temporal coherence must be exploited by tracking keypoints frame to frame.

S5 HOG Descriptors

Divide detection window into 8×8 cell blocks; compute gradient orientation histogram (9 bins, 0–180°) per cell; normalise over 2×2 block (L2-Hys). HOG captures shape and silhouette of traffic signs robustly. Unlike SIFT, HOG operates on a dense grid — crucial for detecting standardised sign shapes even in low contrast. HOG + linear SVM is the classic pedestrian/sign detector (Dalal & Triggs 2005).

4. MATHEMATICAL PRINCIPLES

PHASE CONGRUENCY

$$PC(x) = \Sigma_n W(x)\ \lfloor A_n(x)(\cos(\varphi_n(x)-\bar{\varphi}(x)) - |\sin(\varphi_n(x)-\bar{\varphi}(x))|) - T \rfloor_+ / (\Sigma_n A_n(x) + \varepsilon)$$

A_n = amplitude of n-th Fourier component at scale n

$\bar{\varphi}(x)$ = weighted mean phase

STANDARD HOUGH TRANSFORM (FOR PPHT BASE)

$\rho = x\cdot\cos(\theta) + y\cdot\sin(\theta)$ (line parameterisation)

Accumulator $H(\rho,\theta)$ += 1 for each edge point

PPHT: samples subset, finds peaks, extracts segments

HOG GRADIENT & HISTOGRAM

For each pixel (x,y): $\theta = \text{atan2}(G_y, G_x)$, $m = \sqrt{G_x^2+G_y^2}$

Bin index b = floor(θ/20°) [9 bins for 0-180°]

Block normalisation: $v_{norm} = v / \sqrt{||v||^2 + \varepsilon^2}$

5. COMPARISON: CANNY VS. LOG VS. PHASE CONGRUENCY

PROPERTY	CANNY	LOG	PHASE CONGRUENCY
Illumination Invariance	No (gradient-based)	Partial	**Yes** (phase-based)
Performance in Fog	Poor (low gradients)	Poor	**Good**
Scale Invariance	Partial (σ tuning)	With multi-scale	Built-in (multi-scale Fourier)
Computational Cost	Low	Moderate	High (FFT-based)

ADVANTAGES

- Phase Congruency works where Canny fails in fog
- AGAST 2× faster than FAST (real-time critical)
- HOG robust sign descriptor (standardised shapes)
- PPHT fast lane line extraction

LIMITATIONS

- Phase Congruency: high FFT cost (slow for embedded)
- HOG fixed window — misses multi-scale signs (needs pyramid)
- PPHT probabilistic — may miss faint lane markings
- Heavy rain streaks still confuse keypoints

Q12 Interactive segmentation pipeline for organ extraction from noisy ultrasound volumes (fuzzy boundaries)

1. PROBLEM & TECHNIQUE SELECTION

Ultrasound is inherently noisy (speckle), has poorly defined organ boundaries (acoustic impedance mismatch varies), and suffers from shadowing artefacts. Pipeline: **Marker-Controlled Watershed → Active Contours (Snakes) → Graph-Cut**, in ascending order of refinement.

2. PIPELINE

Ultrasound 3-D Volume → Speckle Filtering (NLM) → Marker-Ctrl Watershed → Active Contour (Snakes / GVF) → Graph-Cut Refinement → Shape Analysis

3. STAGE EXPLANATIONS

S1 Non-Local Means (NLM) Speckle Filtering

Replaces each pixel value with a weighted average of similar-appearing patches across the image: $\hat{u}(p) = \Sigma_q\ w(p,q)\cdot u(q)$, $w(p,q) \propto \exp(-||\text{patch}(p)-\text{patch}(q)||^2/h^2)$. Removes speckle while preserving organ boundary sharpness, unlike Gaussian smoothing which blurs boundaries.

S2 Marker-Controlled Watershed

User places markers inside the organ and in the background. Watershed flooding from these markers creates bounded catchment basins that prevent over-segmentation. Controls the #regions = #markers, preventing the thousands of regions standard watershed would produce on noisy ultrasound. Produces an over-estimate of organ boundary.

S3 Active Contours with GVF (Gradient Vector Flow)

Initialise snake on Watershed boundary. GVF extends the edge-attraction field deep into homogeneous regions: it solves $\partial u/\partial t = \mu\nabla^2 u - (u-\nabla|\nabla I|)\cdot(\nabla|\nabla I|)^2$, where u(x,y) is the flow field. This allows the snake to reach into concave organ boundaries (e.g. kidney hilum) that classical Snakes cannot reach. The snake energy: $E = \alpha|v_s|^2/2 + \beta|v_{ss}|^2/2 - \gamma|\text{GVF}(v)|$.

S4 Graph-Cut Final Refinement

The Snake boundary initialises the foreground/background appearance models. Graph-Cut then globally optimises the binary labelling, correcting errors at shadow artefact regions where the Snake may have leaked. User scribbles on shadow regions provide additional background constraints.

4. MATHEMATICAL PRINCIPLES

SNAKE ENERGY FUNCTIONAL

$E_{snake} = \int_0^1 [\alpha|v_s|^2 + \beta|v_{ss}|^2 - \gamma|\nabla I(v)|^2]\ ds$

α: elasticity, β: bending rigidity, γ: edge-attraction weight

Euler-Lagrange: $\alpha v_{ss} - \beta v_{ssss} - \gamma\nabla(|\nabla I|^2) = 0$

GVF DIFFUSION EQUATIONS

$u_t = \mu(u_{xx}+u_{yy}) - (u-f_x)(f_x^2+f_y^2)$

$v_t = \mu(v_{xx}+v_{yy}) - (v-f_y)(f_x^2+f_y^2)$

$f = |\nabla I|$, (u,v) = GVF field, μ = regularisation weight

ADVANTAGES

- Marker-Watershed prevents over-segmentation
- GVF-Snake handles concave boundaries
- Graph-Cut handles acoustic shadow artefacts
- Fully interactive — user corrections at each stage
- Topologically correct mask for shape analysis

LIMITATIONS

- NLM is computationally heavy (patch search)
- Snake: discrete approximation sensitive to initialisation
- GVF parameter μ needs tuning
- Graph-Cut 3-D: high memory for large ultrasound volumes

Q13 Hybrid anomaly-detection framework for high-dimensional network-traffic feature vectors

1. PROBLEM & TECHNIQUE SELECTION

Network traffic has *extremely rare attack examples*, *high dimensionality* (100+ features), and *concept drift* (new attack types). Pipeline: **Spectral Clustering (natural groupings) → Autoencoder (semi-supervised anomaly via reconstruction error)**.

2. PIPELINE

Traffic Feature Stream → Standardise + PCA(50D) → Spectral Clustering → Autoencoder Training (normal) → Reconstruction Error Threshold → Attack Alert

3. STAGE EXPLANATIONS

S1 Spectral Clustering

Constructs affinity matrix W where $W_{ij} = \exp(-||x_i - x_j||^2/2\sigma^2)$. Graph Laplacian L = D − W; compute top k eigenvectors of L. Cluster in the eigenspace using K-Means. Spectral Clustering finds non-convex clusters (e.g. ring-shaped DDoS traffic patterns) that K-Medoids (Euclidean-based) cannot. Groups traffic into: normal HTTP, DNS, FTP, and potentially anomalous clusters.

S2 Autoencoder (Reconstruction Error)

Architecture: encoder $h = \sigma(Wx+b)$, decoder $\hat{x} = \sigma(W'h+b')$. Trained only on normal traffic samples to minimise $||x-\hat{x}||^2$. Normal traffic → low reconstruction error. Attack traffic (unseen patterns) → high reconstruction error. Decision: flag as attack if $||x-\hat{x}||^2 > \tau$ (threshold set at 99th percentile of normal training error). Robust to unknown attack types (no attack labels needed).

S3 Online Updating

Confirmed normal flows (low error, no alert) retrain the autoencoder via mini-batch SGD every T minutes. The spectral clustering affinity matrix is updated with an exponential moving average of recent pairwise distances, adapting to new normal protocol versions.

4. MATHEMATICAL PRINCIPLES

SPECTRAL CLUSTERING — NORMALISED LAPLACIAN

$$L_{sym} = D^{-1/2}(D-W)D^{-1/2}$$

Eigenvectors $u_1,\ldots,u_k$ of L_{sym} → embed to $\mathbb{R}^k$ → K-Means

$$W_{ij} = \exp(-||x_i-x_j||^2/2\sigma^2)$$

AUTOENCODER RECONSTRUCTION ERROR

$$L(x) = ||x - Dec(Enc(x))||^2_F$$

Anomaly: $L(x) > \tau$, $\tau = \mu_{train} + 3\cdot\sigma_{train}$ (3-sigma rule)

Or: τ = percentile_99 of $\{L(x_i) : x_i \in$ normal train set$\}$

ADVANTAGES

- Spectral clustering captures non-convex attack clusters
- Autoencoder works with zero attack labels
- Detects novel (zero-day) attack types
- Online updating handles concept drift
- Reconstruction error is interpretable

LIMITATIONS

- Spectral clustering $O(n^3)$ affinity matrix — bottleneck for streaming
- Autoencoder may reconstruct some attack patterns if they resemble normal (adversarial attacks)
- Threshold τ requires periodic recalibration
- Deep autoencoder training needs GPU and significant normal data

Q14 Dimensionality-reduction + motion-analysis pipeline for multi-spectral image sequences and object tracking

1. PROBLEM & TECHNIQUE SELECTION

Multi-spectral sequences have many bands with different noise characteristics per band. Tracking objects across frames requires both appearance discrimination and motion estimation. Pipeline: **LDA (supervised class-discriminative reduction) → Parzen Window KDE → MOG2 Background Subtraction + Farneback Dense Optical Flow**.

2. PIPELINE

Multi-spectral Frame Sequence → LDA Reduction → Parzen Window KDE → MOG2 BG Subtraction → Farneback Dense Flow → Object Trajectories

3. STAGE EXPLANATIONS

S1 Linear Discriminant Analysis (LDA)

Finds the d = C−1 linear projections that maximise between-class scatter S_B while minimising within-class scatter S_W: $W = \text{argmax } |W^T S_B W| / |W^T S_W W|$. Unlike PCA (maximises total variance), LDA explicitly preserves class-discriminative information between object classes (vehicles, pedestrians, vegetation). This improves both MOG2 segmentation (class-specific background models) and KDE density estimation. Solves generalised eigenvalue problem: $S_B \cdot w = \lambda \cdot S_W \cdot w$.

S2 Parzen Window KDE (Non-parametric Density)

In the LDA subspace, models the appearance distribution of each object class non-parametrically. Used for object likelihood computation during tracking: $P(\text{class}=k \mid \text{observation}) \propto \hat{p}_k(x)$. The LDA subspace makes KDE effective because it concentrates class-specific density, reducing bandwidth requirements.

S3 MOG2 Background Subtraction in LDA Space

Background model in the compact LDA space requires fewer Gaussian components per pixel. Foreground detection is class-aware: spectral signatures of vehicles differ strongly from vegetation in the LDA projection, reducing false positives.

S4 Farneback Dense Optical Flow

Computed on the LDA-compressed representation between frames t and t+1. Polynomial expansion approximates the local signal as a quadratic, and the displacement field is estimated by equating polynomials after motion. Dense flow gives a velocity vector at every foreground pixel, enabling accurate trajectory estimation for all tracked objects.

4. MATHEMATICAL PRINCIPLES

LDA FISHER CRITERION

$$J(W) = |W^T S_B W| \;/\; |W^T S_W W|$$

$$S_B = \Sigma_c n_x(\mu_c - \mu)(\mu_c - \mu)^T$$

$$S_W = \Sigma_c \Sigma_{x \in Xc} (x - \mu_c)(x - \mu_c)^T$$

Solution: $S_W^{-1} S_B \cdot w = \lambda \cdot w$, keep top C−1 eigenvectors

FARNEBACK POLYNOMIAL EXPANSION

$$f(x) \approx x^T A(x) x + b(x)^T x + c(x)$$

After displacement d: $f(x-d) \approx x^T A' x + (b+2A'd)^T x + \ldots$

Equating: $b + 2A'd = b' \rightarrow d = -A'^{-1}(b'-b)/2$

ADVANTAGES

- LDA maximises class separability (better than PCA for tracking)
- Compact subspace reduces MOG2 and KDE computation
- Dense optical flow — every pixel tracked
- NMF alternative: parts-based (useful for vehicle part tracking)

LIMITATIONS

- LDA requires class labels (supervised) — costly to obtain
- LDA assumes Gaussian class distributions
- Dense Farneback flow is compute-heavy (GPU recommended)
- Brightness constancy violated in multi-spectral band changes

Q15 Hybrid feature extraction for oriented microstructures in SEM/TEM images

⚙ 1. PROBLEM & TECHNIQUE SELECTION

SEM/TEM images contain grain boundaries (lines), crystal lattice patterns (periodic textures), nanoparticle clusters (blobs), and significant electron beam shot noise. Multi-scale and multi-orientation coverage is essential. Pipeline: **Scale-Space Pyramids + Gaussian Derivatives → Log-Gabor Filters → Curvelet Transform → SIFT**.

→ 2. PIPELINE

SEM/TEM Image → Noise Suppression (BM3D) → Gaussian Pyramid + Derivatives → Log-Gabor Filter Bank → Curvelet Transform → SIFT Keypoints → Phase + Texture Classifier

📋 3. STAGE EXPLANATIONS

S1 BM3D Noise Suppression

Block-Matching 3D filtering: groups similar patches into 3-D arrays, applies 3-D transform (wavelet), thresholds, reconstructs. Preserves grain boundary sharpness better than Gaussian smoothing. Electron shot noise in TEM is approximately Poisson-distributed, and BM3D handles this effectively.

S2 Gaussian Pyramid + Derivative Filters

Gaussian derivatives $\partial G/\partial x$, $\partial G/\partial y$, $\partial^2 G/\partial x^2$, $\partial^2 G/\partial x \partial y$ at scales σ, 2σ, 4σ. These form a complete set of local jet descriptors. Second derivatives (Hessian eigenvalues) detect ridge/valley structures = grain boundaries. LoG detects nanoparticles as blobs. All scale-dependent: fine scales capture lattice planes; coarse scales capture grain structure.

S3 Log-Gabor Filter Bank

Log-Gabor avoids the DC component issue of standard Gabor (standard Gabor has a non-zero mean if the bandwidth is large): it is symmetric on a log-frequency axis. Ideal for bandpass filtering: $g(f) = \exp(-(\log(f/f_0))^2/2(\log(\sigma_f/f_0))^2)$. Bank of 6 orientations × 4 frequencies captures grain boundary orientations and periodic lattice spacings in crystal materials with high sensitivity. Avoids the artificial constraint of even-symmetric Gabor at high bandwidth.

S4 Curvelet Transform

Curvelets are a multi-scale representation optimally suited to represent *curve-like singularities* (grain boundaries, dislocation lines). Unlike wavelets (isotropic), curvelets are directional at each scale with anisotropy ratio = 2^{-j/2} at scale j. This makes curvelet coefficients sparse for grain boundary images: few large coefficients → efficient boundary encoding. Replaces DWT when structures are elongated/curved rather than axis-aligned.

S5 SIFT Keypoints

Detects reproducible feature points at nanoparticle locations, lattice defects, and grain triple junctions. SIFT descriptors enable cross-sample matching: identifying the same grain structure across multiple TEM images of different sample regions, supporting statistical grain analysis.

Σ 4. MATHEMATICAL PRINCIPLES

LOG-GABOR FREQUENCY RESPONSE

$$G(f,\theta) = \exp(-(\log(f/f_0))^2/2\sigma_r^2) \cdot \exp(-(\theta-\theta_0)^2/2\sigma_\theta^2)$$

f_0: centre frequency, σ_r: bandwidth on log scale

Advantage: symmetric bandwidth, no DC component

CURVELET ANISOTROPY SCALING

Curvelet at scale j, direction ℓ: $\psi_{j,\ell}(x)$

Support size: 2^{-j} × 2^{-j/2} (anisotropic: width² ∝ length)

Parabolic scaling law: optimal for C^2 curves (Candès & Donoho 2004)

N-term approximation error: $||f - f_N||^2 \leq C \cdot (\log N)^3 \cdot N^{-2}$ vs. wavelet N^{-1}

GAUSSIAN DERIVATIVE FILTER BANK (JET)

$J^n(x;\sigma) = G(x;\sigma) * \partial^n I/\partial x^n \partial y^m$, $n+m \leq N$

Local jet descriptor: J = [G, G_x, G_y, G_xx, G_xy, G_yy, …] at each scale σ

✓ 5. COMPARISON VS. CANNY / HARRIS / HOUGH

PROPERTY	HYBRID PROPOSED	CANNY	HARRIS	HOUGH
Curved Boundary Handling	**Excellent** (Curvelet)	Poor	No	Straight lines only
Periodic Texture	**Yes** (Log-Gabor)	No	No	No
Scale Invariance	**Yes** (pyramid + SIFT)	No	No	No
Noise Resilience	**High** (BM3D + Curvelet)	Low	Low	Moderate
Grain Boundary Sparsity	**Yes** (Curvelet sparse)	Dense edges	Only corners	Only lines

ADVANTAGES

- Curvelet sparse representation of grain boundaries
- Log-Gabor handles broad bandwidth lattice frequencies
- Multi-scale covers atomic-scale to grain-scale
- SIFT enables cross-image grain matching
- BM3D preserves sharp boundaries despite heavy noise

LIMITATIONS

- Curvelet transform is computationally expensive (wrapping implementation needed)
- Log-Gabor frequency centre f_0 and bandwidth need domain calibration (TEM accelerating voltage affects lattice spacing)
- SIFT may miss sub-pixel lattice defects
- High-dimensional feature vector requires supervised dimensionality reduction (LDA) before classification
- BM3D assumes stationary noise — TEM dose rate variation violates this